Total No. of Questions : 4]

SEAT No. :

PF229

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APR-26/SE/Insem-283

S.E. (AIML) (Insem)

DATABASE MANAGEMENT SYSTEM

(2019 Pattern) (Semester - IV) (218554)

Time : 1 Hour] ***[Max. Marks : 30***

Instructions to the candidates:

1) ***Answer Q.1 or Q.2 and Q.3 or Q.4.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume suitable data, if necessary.***

Q1) a) Explain the components of a DBMS with neat diagram. **[6]**

b) List four significant differences between a file-processing system and DBMS. **[4]**

c) List and Explain different type of attribute in ER data model with example. **[5]**

OR

Q2) a) What is a data model? Discuss any two types of data model. **[5]**

b) Describe the three levels of data abstraction with a neat diagram. **[5]**

c) Discuss the concept of data independence and explain its importance in a database environment. **[5]**

Q3) a) Discuss the differences between the candidate keys and the primary key of a relation. Explain what is meant by a foreign key. How do foreign keys of relations relate to candidate keys? **[5]**

b) What is a view? Discuss the difference between a view and a base relation. Explain with example. **[5]**

c) List and explain any 6 Codd's rule. **[5]**

OR

Q4) a) Construct an E-R diagram for a car insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents. Each insurance policy covers one or more cars, and has one or more premium payments associated with it. Each payment is for a particular period of time and has an associated due date and the date when the payment was received. **[5]**

b) Explain properties (any three) of Relations. Describe the terms relation and relation schema. **[5]**

c) Define the two principal integrity rules for the relational model. Discuss why it is desirable to enforce these rules. **[5]**

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